

Papers-to-Table

The Architecture of Auditable AI Extraction

Local-first schema extraction • Human-in-the-loop review • Immutable provenance

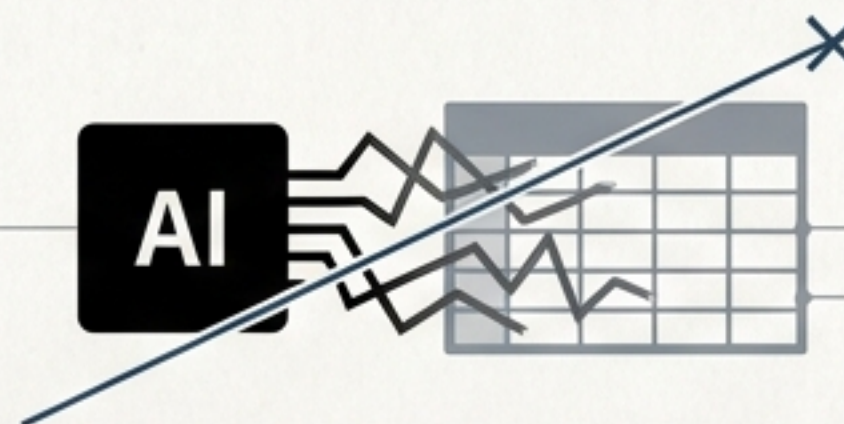
Setting the Baseline: AI-Assisted Review

AI-Assisted Review (What It Is)



- Schema-first extraction based on explicit column definitions
- Proposes one best value per cell with attached, verifiable evidence
- Human reviewer explicitly decides what to accept or reject
- Writes approved changes to a completely new, separate export file

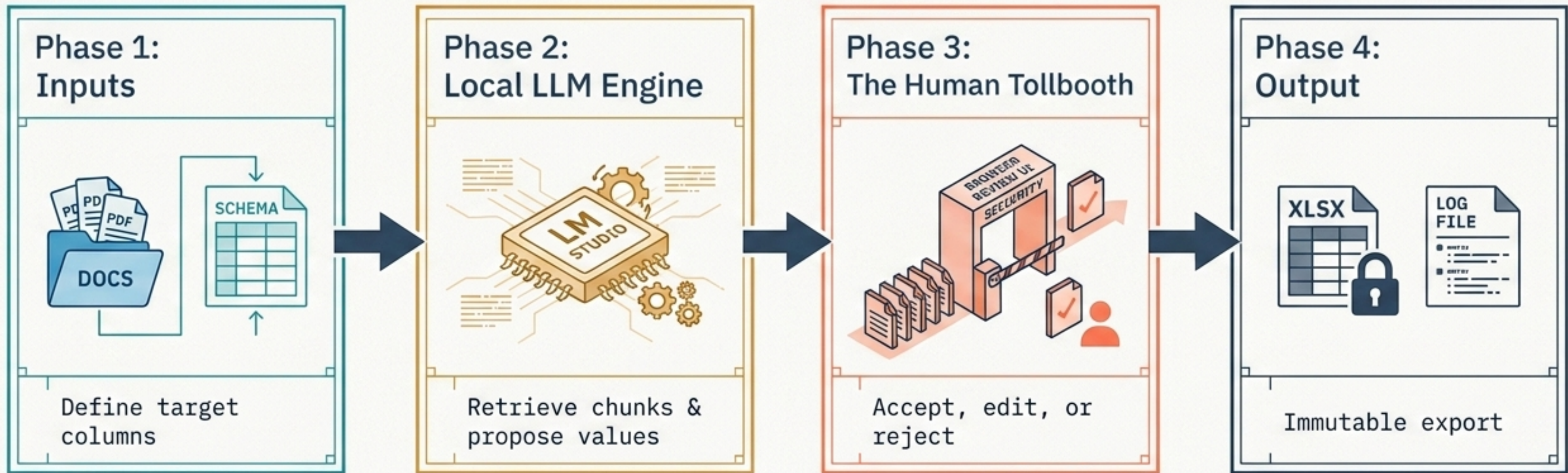
Autonomous AI (What It Isn't)



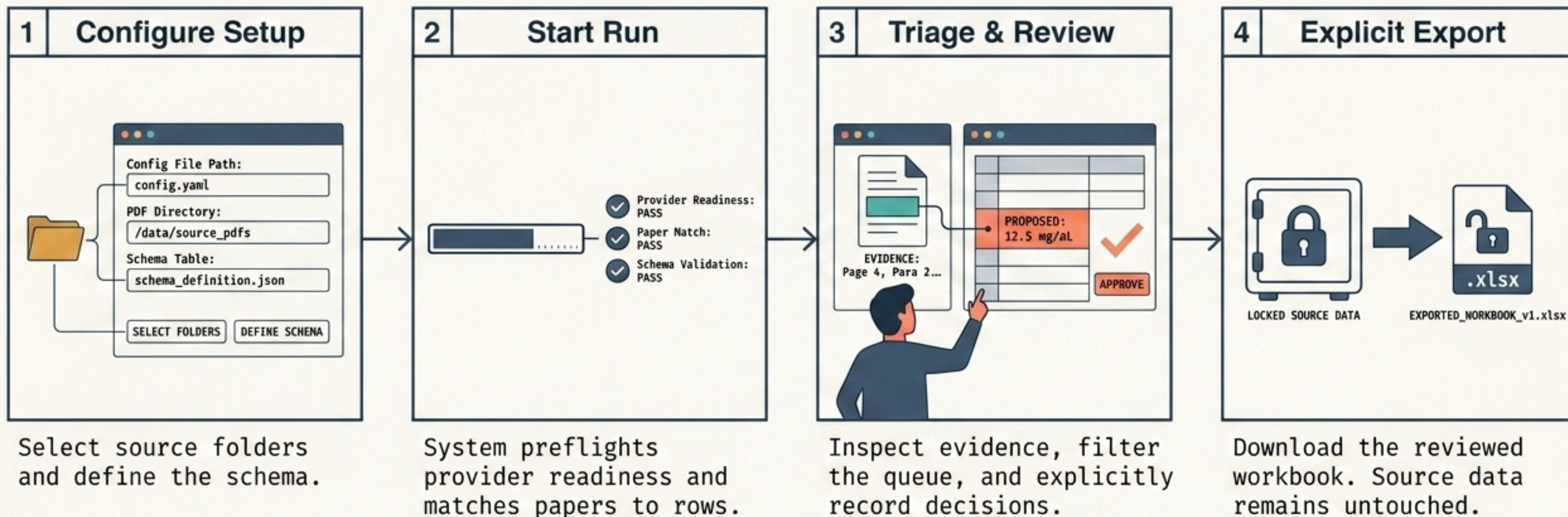
- Broad, conversational chatbot-like interactions
- Unreviewed AI guesses exported automatically to end users
- Direct, in-place mutation and overwriting of the source spreadsheet

The system does not replace expert scientific judgment; it accelerates the curation process by tethering proposed facts directly to source evidence.

The Extraction Lifecycle



The Operator Workflow

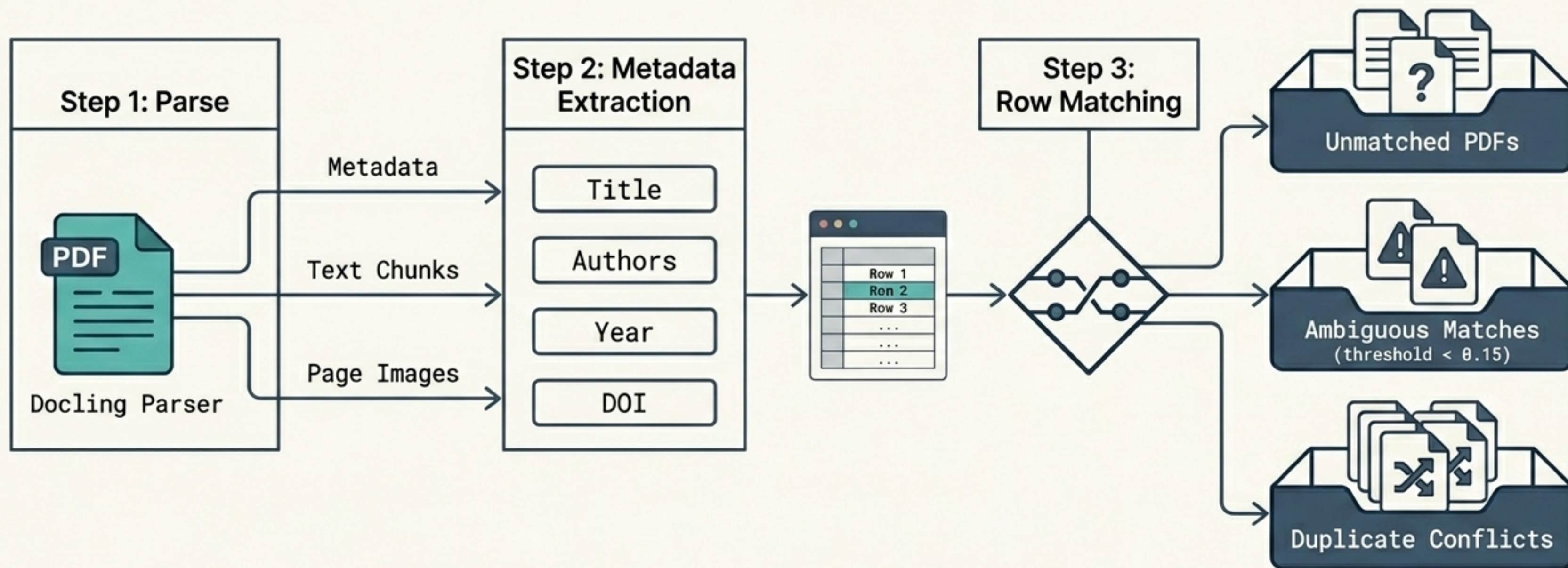


Operational Context: Run Modes

	Normal Mode	Verify Mode	Eval Mode
Primary Use Case	Extraction for empty targets.	Reviewer comparison on already-filled cells.	Leakage-aware benchmarking.
Target Cells	Empty.	Filled.	Masked working copy (app-owned).
Gold Data Protection	N/A.	Existing cells act as format helpers.	Strict isolation. Target gold values are never exposed to extraction prompts.

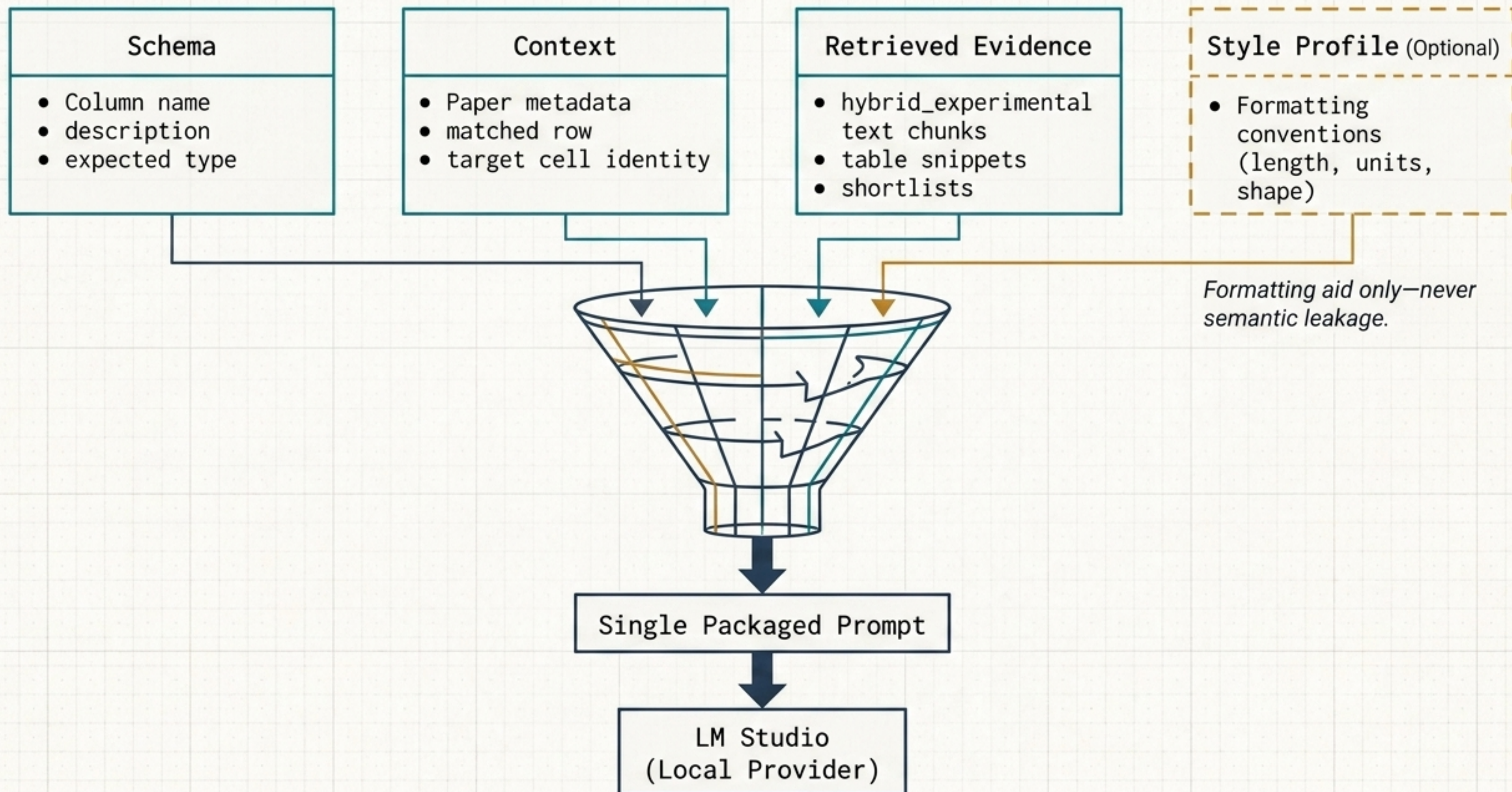
Eval mode ensures reproducible downstream scoring by preserving masked-table provenance.

The Pipeline Phase I: Ingest & Match

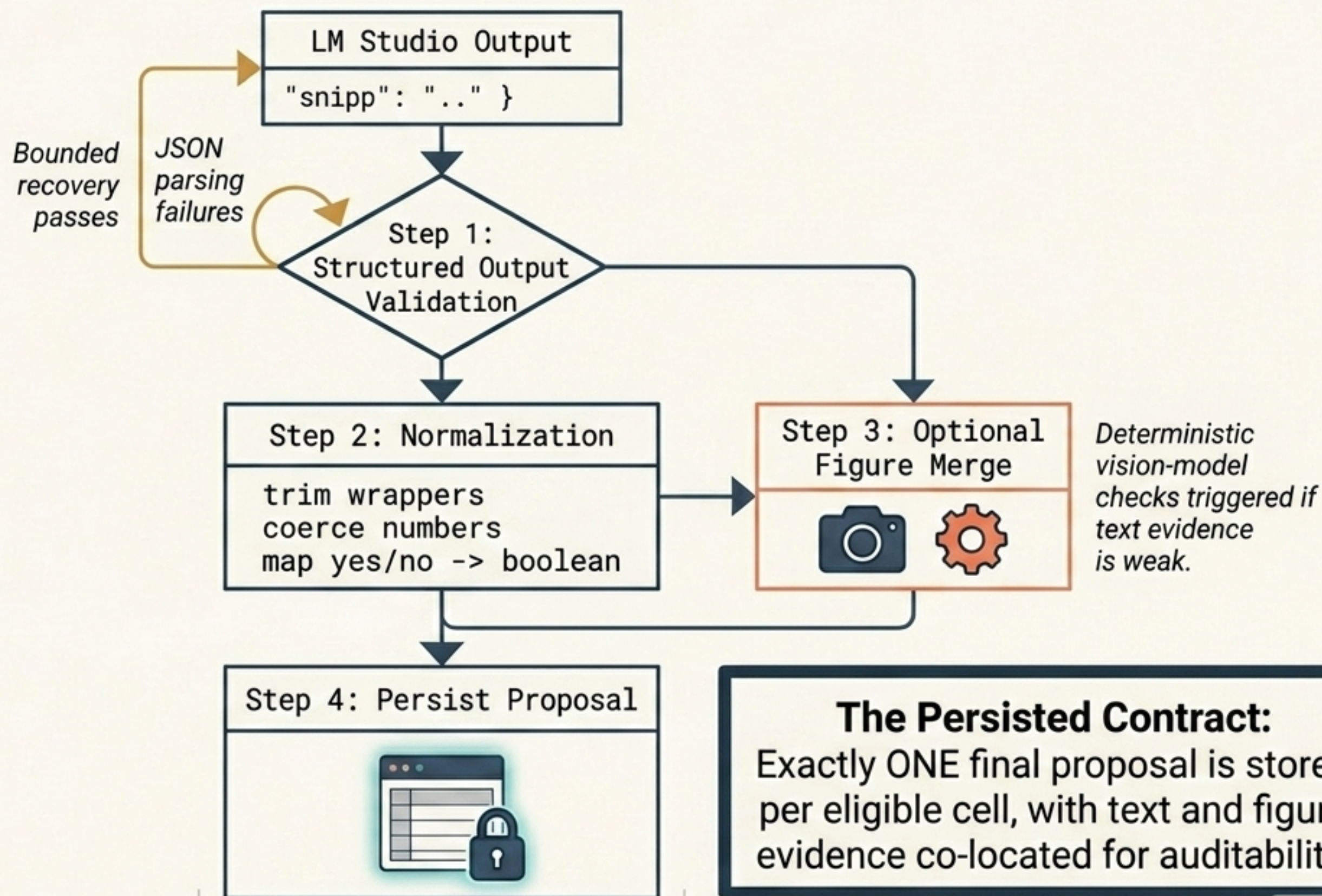


Matching is a distinct stage before proposal generation. Unmatched or ambiguous situations remain explicit, never silently coerced.

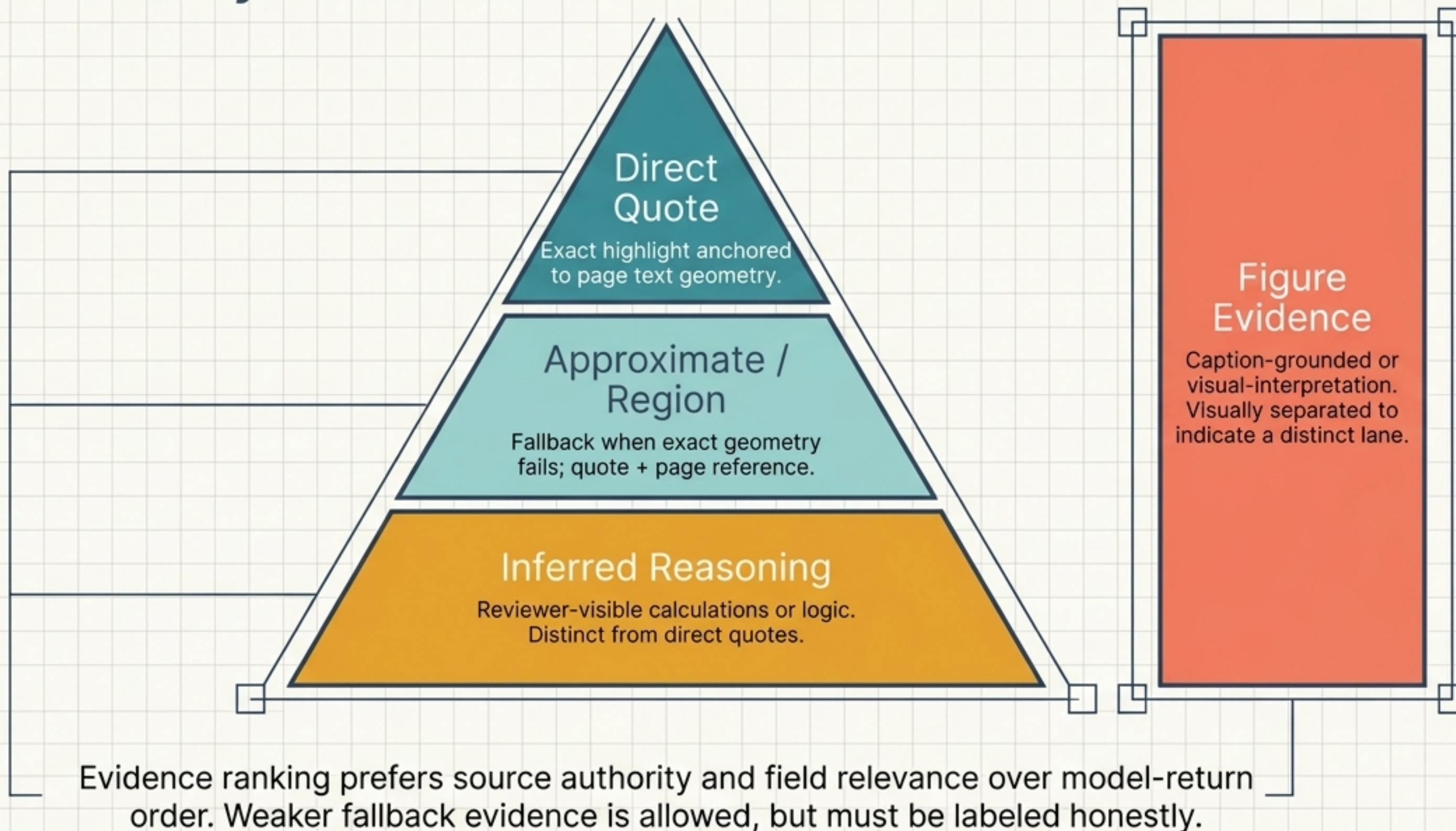
The Pipeline Phase II: Prompt Composition



The Pipeline Phase II: Decide & Normalize



The Anatomy of Evidence

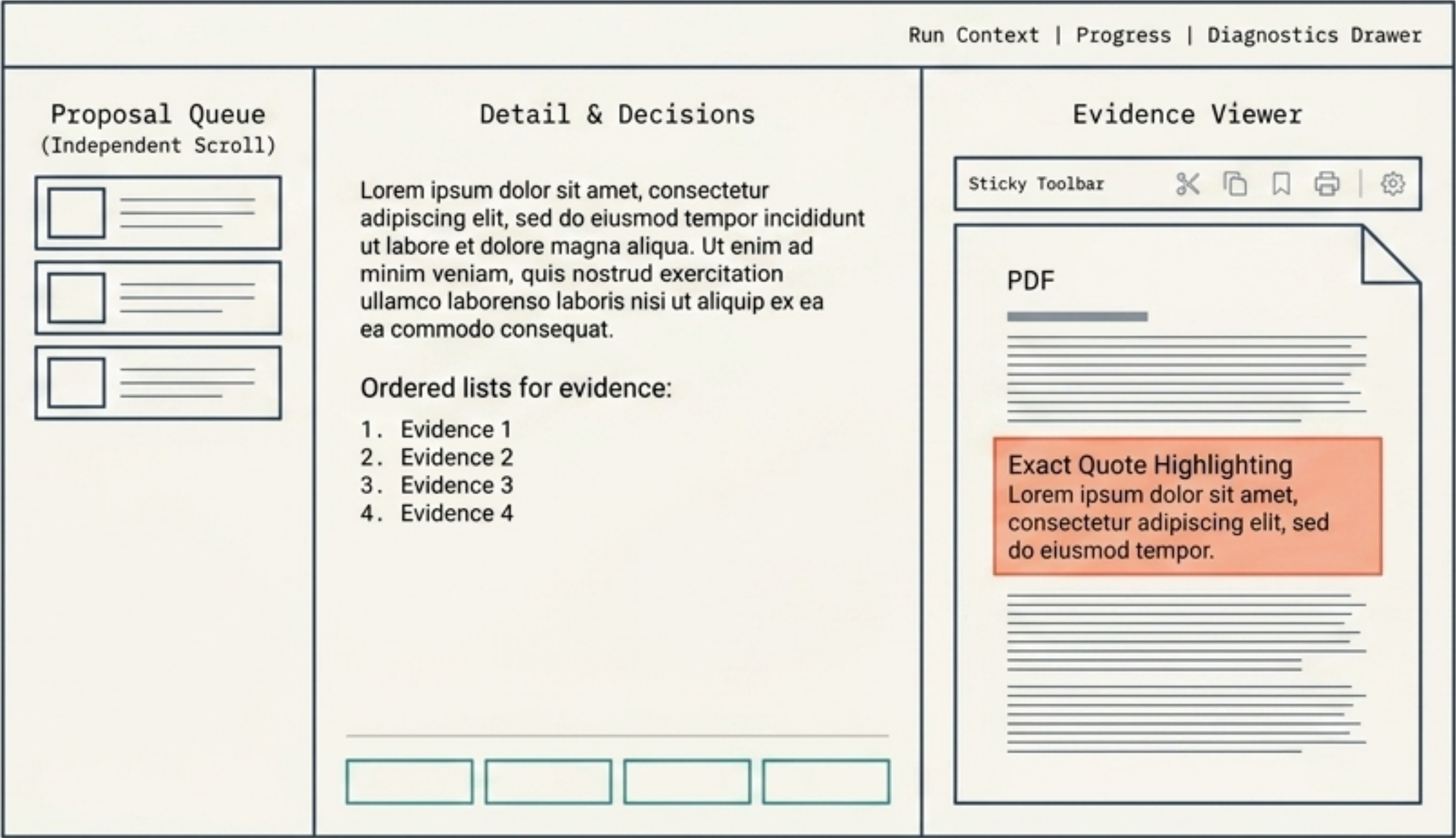


Tethering Facts to Sources



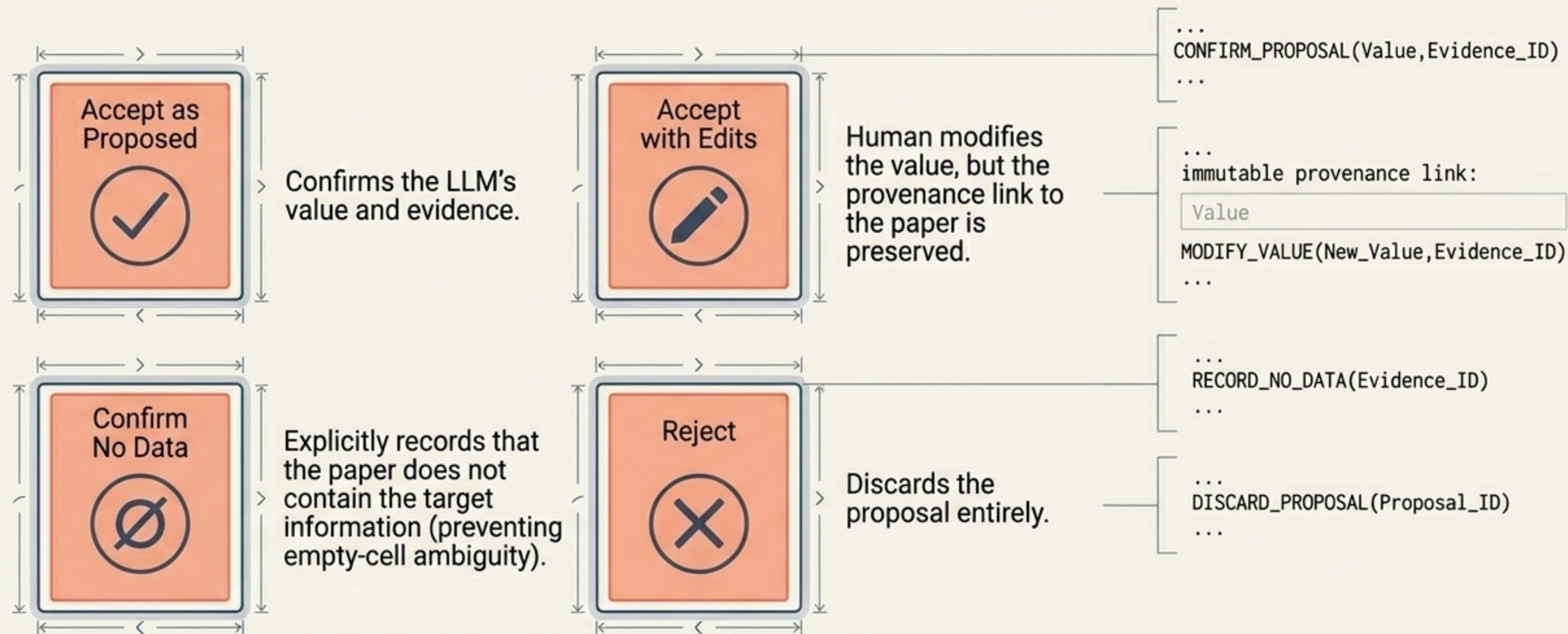
Provenance is immutable. The reviewer sees exactly why a value was proposed without hunting through the document.

The Curation Workspace



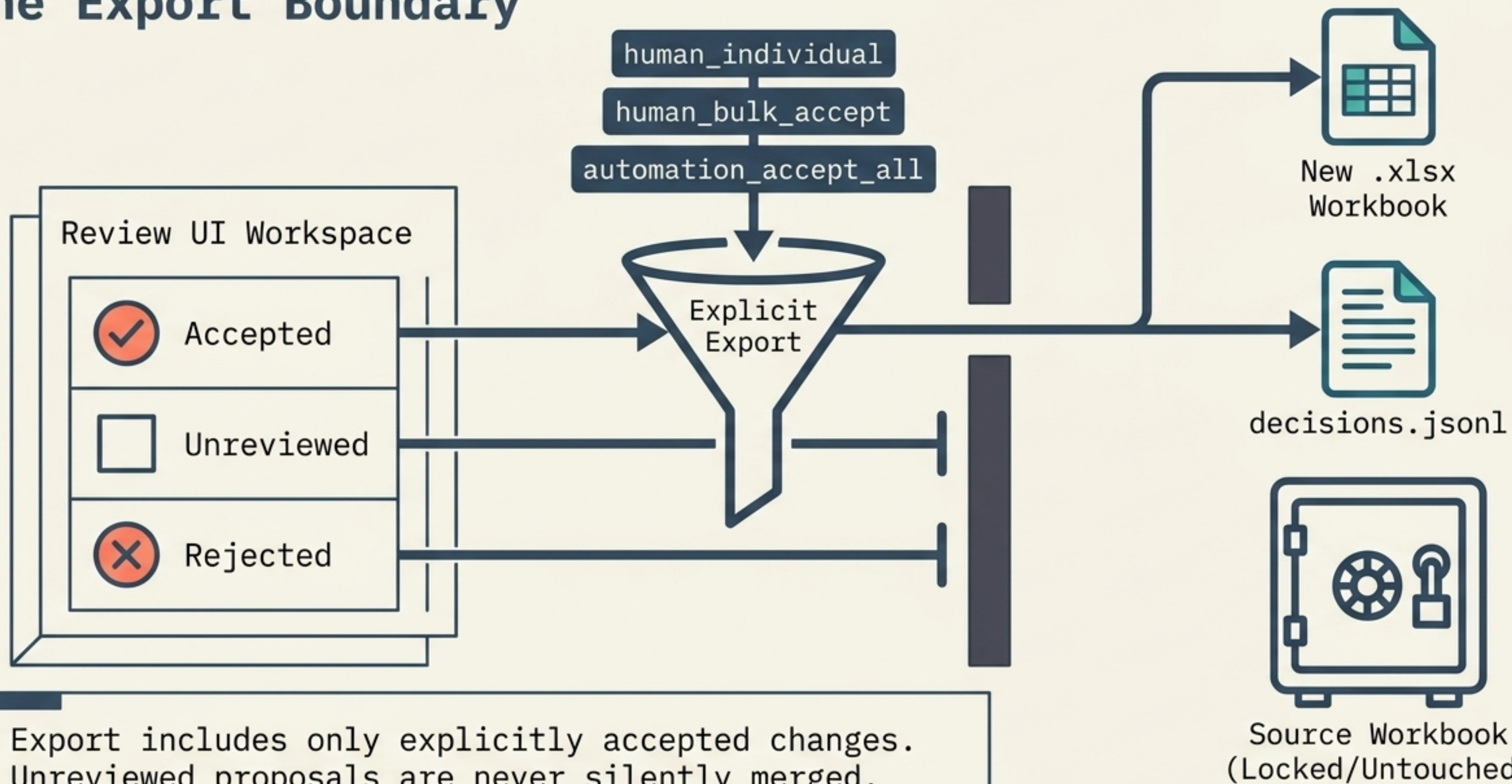
A strictly Queue-First design. Optimized for sustained, focused curation rather than freeform exploration.

First-Class Curation Outcomes

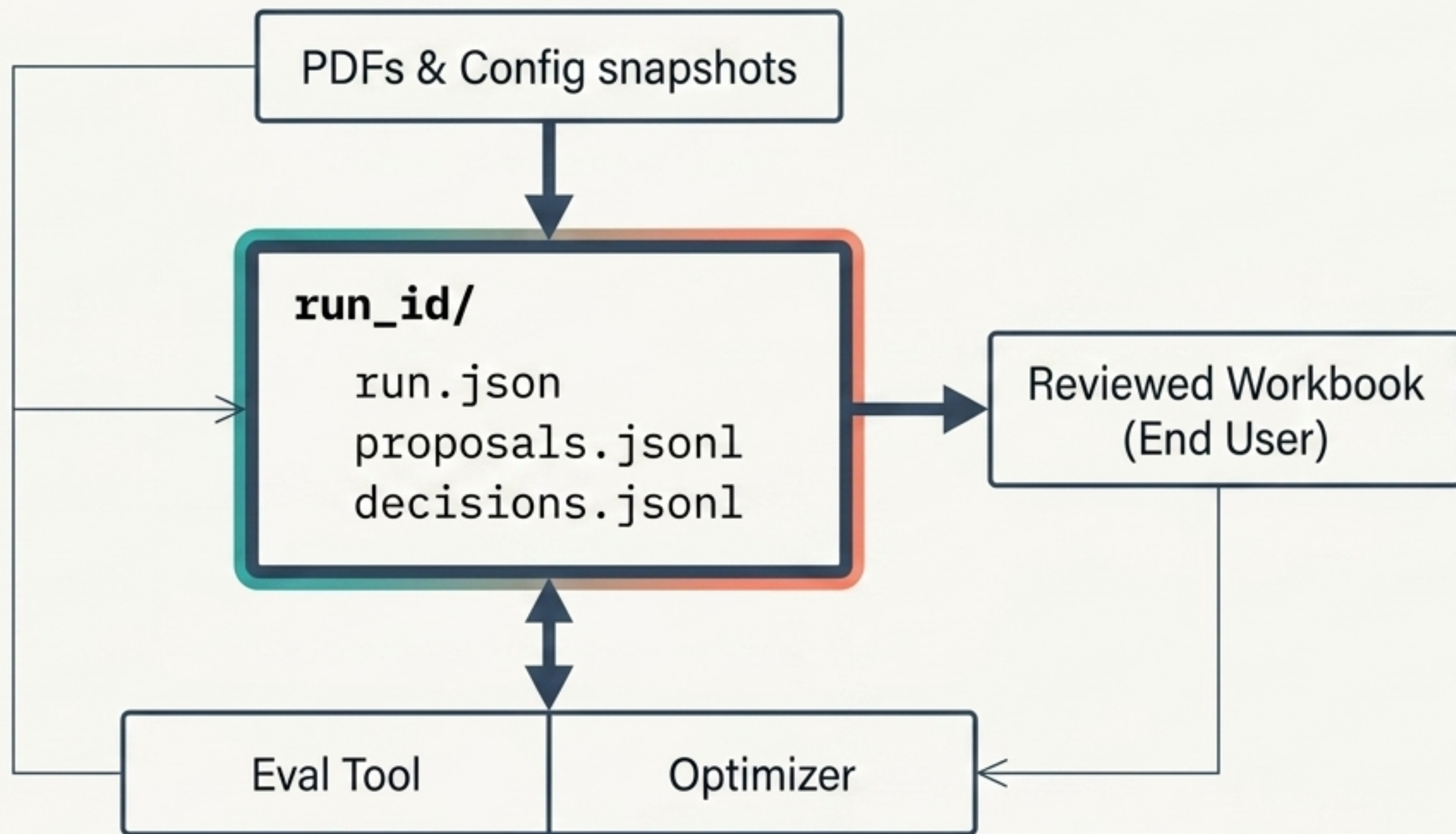


The reviewer is reviewing what the paper supports, not grading the model.
Every decision is a first-class, persisted artifact.

The Export Boundary



Synthesis: The Auditable Run Bundle



The final product is not just a filled spreadsheet. It is the Run Bundle—a complete, filesystem-based artifact that permanently links inputs, AI proposals, human decisions, and downstream benchmarking.

Precision over Guesswork



Human Review

Papers-to-table relies on a fundamental premise: scientific extraction requires evidence, provenance, and human authority.

By separating AI proposal generation from human curation, we ensure that every cell in the final table is explicitly supported, fully auditable, and confidently tethered to its source.